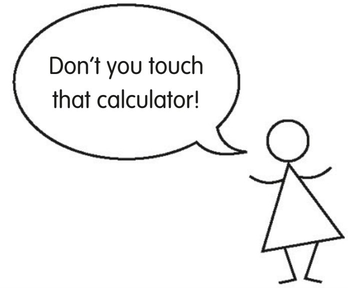
Long multiplication



Multiplying a large number by a single-digit number

This isn't as hard as it looks, but you will need to know your times tables up to **10 × 10** pretty well before you try this.

Write the large number above the small one.

```
  H T U
  7 8 6
×     2
-------
```

Multiply the single-digit number on the bottom by the units, then tens, then hundreds of the number at the top.

Multiply by units

```
  786
×   2
-----
   12 → 6 × 2 = 12
```

Multiply by tens

```
  786
×   2
-----
   12
  160 → 80 × 2 = 160
```

Multiply by hundreds

```
  786
×   2
-----
   12
  160
 1400 → 700 × 2 = 1,400
```

Finally, add up the answers to those three multiplications.

```
    12
   160
+ 1400
------
  1572
```

So: 786 × 2 = 1,572

Now let's try the fast way

A quicker way of doing this is to write the answer to each multiplication on the same line, going from right to left. If you get an answer of ten or more when you're multiplying the units, tens, or hundreds, you "carry" the first digit of that answer, adding it to the column to the left.

```
  285
×   3      5 × 3 = 15
-----
    5
-----
   1       Carry 1 to tens column.
```

```
  285      8 × 3 = 24
×   3
-----
   55      24 + 1 = 25
-----
  2 1      Carry 2 to hundreds column.
```

```
  285      2 × 3 = 6
×   3
-----
  855      6 + 2 = 8
-----
  2 1
```

Now have a go

```
  385      723      210      974
×   2    ×   4    ×   3    ×   8
-----    -----    -----    -----
```


Answers: Blue – 770. Red – 2,892. Green – 630. Yellow – 7,792.